

CIHRM Enterprise HRMS

Product Requirements Document (PRD)

Version: 1.0

Prepared For: CIHRM (Chartered Institute of Human Resource Management, Ghana)

Product Type: Enterprise Web-Based HR Management Platform

Prepared By: Product & System Architecture Team

1. Executive Summary

1.1 Product Overview

The CIHRM Enterprise Human Resource Management System (HRMS) is a modern web-first enterprise platform designed to centralize organizational operations, HR management, internal workflows, employee services, governance, and performance management into a single secure and scalable digital ecosystem.

The platform combines:

- Human Resource Management
- Internal Service Desk Operations
- Workflow Automation
- Performance & KPI Management
- Complaint & Governance Management
- Analytics & Executive Reporting

The system is designed as a modular enterprise application with future scalability toward a multi-organization SaaS platform.

2. Product Vision

To build a modern enterprise digital workplace platform that transforms organizational management through automation, transparency, analytics, and secure operational workflows.

3. Product Goals

3.1 Business Goals

- Digitize HR operations
 - Improve operational efficiency
 - Centralize organizational data
 - Reduce manual workflows
 - Improve governance and accountability
 - Enable data-driven decision-making
 - Improve employee experience
 - Enhance service delivery across departments
 - Position CIHRM as a digitally advanced institution
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3.2 Product Goals

- Deliver a web-first enterprise platform
 - Implement scalable modular architecture
 - Provide enterprise-grade RBAC security
 - Deliver real-time dashboards and analytics
 - Automate workflows and approvals
 - Provide enterprise reporting capabilities
 - Ensure high system availability and reliability
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4. Target Users

4.1 Primary Users

User Type	Description
HR Administrators	Manage employee records and HR workflows
Managers	Approve workflows and monitor teams
Employees	Access self-service HR features
Finance Officers	Manage payroll and compensation
IT Support	Manage tickets and technical support
Executives	Access dashboards and analytics
Auditors	Access compliance and audit reports

5. Product Scope

5.1 In Scope

The following features are included in the initial enterprise platform.

Core HR

- Employee management
- Department management
- Organizational hierarchy
- Staff documents

Attendance & Leave

- Attendance tracking
- Leave requests
- Leave approvals
- Holiday calendar

Payroll

- Salary structures
- Payroll processing
- Payslip generation
- Tax and SSNIT calculations

Performance & KPI

- KPI assignment
- Performance tracking
- Scorecards
- Goal management

Internal Ticketing

- Ticket creation
- SLA management
- Ticket assignment
- Resolution tracking

Complaint Management

- Anonymous reporting
- Investigation workflows
- Resolution tracking

Recruitment

- Job postings
- Applicant tracking
- Interview scheduling

Reports & Analytics

- Executive dashboards
- Department analytics
- Exportable reports

Security & Administration

- RBAC
 - MFA
 - Audit logs
 - Workflow management
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5.2 Out of Scope (Initial Phase)

The following features are excluded from the MVP:

- Native mobile applications
- AI automation engine
- Government API integrations
- Advanced LMS system
- Multi-tenant SaaS support
- External payroll banking integrations

These features are planned for future phases.

6. Product Strategy

6.1 Web-First Strategy

The platform shall prioritize browser-based enterprise accessibility.

Benefits

- Easier deployment
- Lower maintenance costs
- Faster adoption
- Cross-device support

- Real-time centralized updates

Supported Devices

- Desktop browsers
 - Tablets
 - Mobile browsers
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6.2 Modular Architecture Strategy

The system shall be developed as independent business modules connected through centralized authentication and shared enterprise services.

Architectural Benefits

- Easier maintenance
 - Independent feature development
 - Better scalability
 - Reduced complexity
 - Faster upgrades
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7. Technical Stack

7.1 Frontend

Technology	Purpose
Vue.js	Frontend framework
Vue Router	Routing
Pinia	State management
TypeScript	Type safety
Tailwind CSS	Styling framework
PrimeVue / Vuetify	Enterprise UI components
Axios	API communication
Chart.js / ApexCharts	Analytics visualizations

7.2 Backend

Technology	Purpose
Laravel	Backend framework
PHP 8.x	Backend language
Laravel Sanctum	Authentication
Laravel Queues	Background processing
Laravel Horizon	Queue monitoring
Laravel Events	System event handling

7.3 Database

Technology	Purpose
PostgreSQL	Enterprise relational database
Redis	Cache and queues

7.4 Infrastructure

Technology	Purpose
Nginx	Web server
Docker	Containerization
GitHub Actions	CI/CD
Azure Cloud	Cloud hosting

8. System Architecture

8.1 High-Level Architecture

Vue.js Frontend
↓
Laravel REST API

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PostgreSQL Database

8.2 Architectural Layers

Presentation Layer

Responsible for:

- UI rendering
- Dashboards
- User interactions
- Forms and workflows

Application Layer

Responsible for:

- Business logic
- APIs
- Authentication
- Workflow processing
- Notifications

Data Layer

Responsible for:

- Data persistence
 - Transactions
 - Reporting data
 - Audit logs
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9. Core Modules

9.1 Dashboard Module

Description

The dashboard serves as the centralized operational overview for users.

Features

Executive Dashboard

- Organization KPIs
- Staff analytics
- Attendance trends
- Ticket analytics
- Payroll overview
- Department performance

Department Dashboard

- Team performance
- Pending approvals
- Open tickets
- Attendance statistics

Employee Dashboard

- Personal profile
 - Leave balances
 - Assigned tasks
 - Notifications
 - KPI progress
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UI Requirements

- Responsive grid layout
 - Interactive charts
 - KPI cards
 - Notification center
 - Quick actions
 - Activity timeline
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9.2 Employee Management Module

Features

- Employee profiles
- Department assignment
- Employment history
- Digital employee files
- Search and filtering

- Employee status management
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Functional Requirements

ID	Requirement
EMP-001	Create employee profiles
EMP-002	Upload employee documents
EMP-003	Assign departments and roles
EMP-004	Search and filter employees
EMP-005	View employee activity logs

9.3 Attendance & Leave Module

Features

- Clock-in/out tracking
 - Leave requests
 - Leave approvals
 - Leave balances
 - Attendance reports
 - Ghana holiday calendar
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Functional Requirements

ID	Requirement
ATT-001	Record employee attendance
ATT-002	Submit leave requests
ATT-003	Approve/reject leave
ATT-004	Track leave balances
ATT-005	Generate attendance reports

9.4 Payroll Module

Features

- Payroll processing
 - Salary structures
 - Payslip generation
 - Tax calculations
 - SSNIT calculations
 - Payroll reports
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Functional Requirements

ID	Requirement
PAY-001	Configure salary structures
PAY-002	Process payroll
PAY-003	Generate payslips
PAY-004	Calculate deductions
PAY-005	Export payroll reports

9.5 Performance & KPI Module

Features

- KPI assignment
 - Goal tracking
 - OKR management
 - Performance reviews
 - Department scorecards
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Functional Requirements

ID	Requirement
KPI-001	Assign KPIs
KPI-002	Track KPI progress

ID	Requirement
KPI-003	Conduct performance reviews
KPI-004	Generate scorecards
KPI-005	Analyze department performance

9.6 Ticketing / Service Desk Module

Features

- Ticket creation
 - Ticket assignment
 - SLA tracking
 - Ticket escalation
 - Ticket resolution
 - Ticket comments and attachments
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Functional Requirements

ID	Requirement
TKT-001	Create tickets
TKT-002	Assign tickets
TKT-003	Track SLA compliance
TKT-004	Escalate overdue tickets
TKT-005	Close and archive tickets

9.7 Complaints & Governance Module

Features

- Anonymous reporting
 - Case management
 - Investigation tracking
 - Resolution workflows
 - Escalation handling
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Functional Requirements

ID	Requirement
GOV-001	Submit anonymous complaints
GOV-002	Assign investigators
GOV-003	Escalate unresolved cases
GOV-004	Track resolutions
GOV-005	Generate governance reports

9.8 Recruitment Module

Features

- Job postings
 - Applicant tracking
 - CV uploads
 - Interview scheduling
 - Offer management
 - Onboarding workflows
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Functional Requirements

ID	Requirement
REC-001	Create job postings
REC-002	Track applicants
REC-003	Schedule interviews
REC-004	Upload candidate documents
REC-005	Manage onboarding tasks

10. Workflow Engine

10.1 Overview

The platform shall include a configurable workflow automation engine.

Supported Workflow Areas

Leave Approvals

Employee → Manager → HR

Payroll Approvals

Finance Officer → Finance Manager → Executive Approval

Ticket Escalation

Support Agent → Department Lead → Management

Complaint Investigation

Investigator → Governance Committee → Resolution

Workflow Features

- Configurable approvals
 - Escalation rules
 - SLA timers
 - Automated notifications
 - Activity tracking
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11. Role-Based Access Control (RBAC)

11.1 Roles

Role	Permissions
Super Admin	Full system control

Role	Permissions
HR Admin	HR management
Manager	Department approvals
Employee	Self-service access
Finance Officer	Payroll access
IT Support	Ticket management
Auditor	Read-only access

11.2 RBAC Requirements

ID	Requirement
SEC-001	Role-based module access
SEC-002	Department restrictions
SEC-003	Granular permissions
SEC-004	Approval hierarchy enforcement
SEC-005	Audit trail logging

12. Dashboard & Analytics Requirements

12.1 Executive Dashboard

Metrics

- Employee count
- Attendance rate
- Payroll summary
- KPI completion
- Ticket analytics
- Department performance

12.2 Reporting Features

- PDF export
- Excel export

- CSV export
 - Date filtering
 - Department filtering
 - Drill-down reports
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13. Notification System

Supported Channels

- In-app notifications
 - Email notifications
 - SMS notifications
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Notification Triggers

- Leave approvals
 - Ticket assignments
 - Payroll completion
 - KPI deadlines
 - Escalation alerts
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14. Security Requirements

14.1 Authentication

- JWT authentication
 - Secure password policies
 - Session management
 - Refresh tokens
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14.2 Multi-Factor Authentication

Mandatory for:

- HR Admins
 - Finance Officers
 - Executives
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14.3 Audit Logging

Track:

- User activity
 - Login history
 - Data changes
 - Permission changes
 - Workflow actions
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15. Non-Functional Requirements

15.1 Performance

- Fast dashboard loading
 - Optimized API responses
 - Efficient database queries
 - Concurrent user support
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15.2 Scalability

The system shall support:

- Future module expansion
 - Increased user load
 - Multi-organization support
 - Cloud scaling
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15.3 Availability

Target uptime:

- 99.9% availability
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15.4 Usability

The system shall provide:

- Intuitive navigation

- Responsive layouts
 - Accessibility support
 - Minimal learning curve
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16. API Requirements

API Standards

- RESTful architecture
 - JSON responses
 - Secure authentication
 - Versioned APIs
 - Standardized error handling
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Sample API Modules

- /api/employees
 - /api/attendance
 - /api/leave
 - /api/payroll
 - /api/tickets
 - /api/kpis
 - /api/reports
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17. Database Requirements

Core Tables

HR Tables

- employees
- departments
- positions
- employee_documents

Attendance Tables

- attendance
- leave_requests
- leave_balances

Payroll Tables

- payroll_runs
- salary_structures
- payslips

Ticketing Tables

- tickets
- ticket_comments
- sla_policies

KPI Tables

- kpis
- performance_reviews
- goals

Governance Tables

- complaints
 - investigations
 - resolutions
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18. DevOps & Deployment

18.1 CI/CD

Recommended Tools

- GitHub Actions
 - Docker
 - Docker Compose
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18.2 Environment Strategy

Environments

- Development
 - Staging
 - Production
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18.3 Monitoring

Tools

- Azure Monitor
 - Grafana
 - Prometheus
 - ELK Stack
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19. Product Roadmap

Phase 1 – Core Foundation

Duration: 2–3 Months

Modules:

- Authentication
 - RBAC
 - Dashboard framework
 - Employee management
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Phase 2 – HR Operations

Duration: 2 Months

Modules:

- Attendance
 - Leave management
 - Payroll
 - Notifications
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Phase 3 – Organizational Services

Duration: 2 Months

Modules:

- KPI management
 - Ticketing
 - Complaint management
 - Asset management
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Phase 4 – Intelligence & Expansion

Duration: 3 Months

Modules:

- Advanced analytics
 - Mobile optimization
 - AI insights
 - External integrations
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20. Success Metrics

Operational Metrics

- Reduced HR processing time
 - Faster ticket resolution
 - Reduced manual workflows
 - Improved attendance tracking
 - Increased workflow automation
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Business Metrics

- Increased operational efficiency
 - Improved employee satisfaction
 - Better organizational transparency
 - Improved reporting accuracy
 - Faster executive decision-making
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21. Risks & Dependencies

Risks

- Payroll compliance complexity
 - User adoption resistance
 - Workflow configuration complexity
 - Security and data privacy concerns
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Dependencies

- Cloud infrastructure setup
 - Department process mapping
 - Organizational hierarchy definition
 - User training and onboarding
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22. Conclusion

The CIHRM Enterprise HRMS is designed as a modern, scalable, secure, and workflow-driven enterprise platform that centralizes HR operations, organizational services, analytics, and governance into a single digital ecosystem.

The platform positions CIHRM as a technologically advanced institution capable of leading enterprise HR transformation across Ghana and Africa.